

IE 492 — Senior Design Project · Final Report

Stop the Fire!

*A network optimization approach to wildfire
containment in the Köyceğiz–Marmaris region*

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Abstract

Wildfires in the eastern Mediterranean have become more frequent and more destructive over the past decade, and the Turkish forestry agency (OGM) still relies on a largely reactive operational doctrine: detect a fire, dispatch ground crews, and try to suppress the flame front after it has already started to expand. This project builds a proactive, graph-theoretic decision support tool that recommends *which* parts of a real forest landscape a small number of intervention units should defend first, before the front reaches them. We model the landscape as a planar graph extracted from satellite land-cover data and open street/river maps, simulate fire propagation as a discrete deterministic process on this graph, and compare seven principled containment strategies plus a random baseline against each other on 4,752 controlled synthetic runs and a real Köyceğiz–Marmaris case study. The strategies fall under three philosophies: local greedy heuristics (which protect the vertex that looks best right now), global structural heuristics (which look for natural bottlenecks in the graph), and a one-step lookahead policy (which simulates the engine inside its own decision-making). On a 4,752-run synthetic Delaunay benchmark, the one-step lookahead policy burns 27.5% of the network on average versus 70.1% for random selection, a 43-point gap, while running in roughly 2 ms per turn. On the real Köyceğiz map (101 vertices, 50,280 ha of forest, 172 active edges, $k = 2$ firefighters per turn), the same policy contains the fire to 16.8% of the network. The full pipeline is wrapped in a web application (FastAPI backend, React/Leaflet frontend) that lets a forestry planner draw a bounding box on a map, build the corresponding graph in roughly a second, pick a fire origin, and read off an ordered list of vertices to defend, together with a recommendation card explaining why a particular strategy was chosen for the topology in question.

Keywords: Firefighter Problem, wildfire containment, network optimization, planar graphs, min-cut, lookahead heuristics, decision support, GIS, ESA WorldCover.

Özet

Doğu Akdeniz’de orman yangınları son on yılda hem sıklık hem de tahribat açısından belirgin biçimde artmıştır; buna karşın Orman Genel Müdürlüğü’nün (OGM) saha doktrini büyük ölçüde tepkiseldir: yangın tespit edilir, yer ekipleri sevk edilir ve alev cephesi yayılmaya başladıktan sonra söndürülmeye çalışılır. Bu proje, küçük sayıda müdahale biriminin gerçek bir orman örüntüsünün *hangi* bölümlerini, cephe oraya ulaşmadan önce öncelikli olarak savunması gerektiğini öneren proaktif ve çizge-kuramsal bir karar destek aracı geliştirir. Orman örüntüsü, uydu arazi-örtüsü verilerinden ve açık yol/akarsu haritalarından çıkarılan bir düzlemsel çizge olarak modellenir; yangın yayılımı bu çizge üzerinde ayrık ve determinist bir süreç olarak benzetilir; yedi ilkeli sınırlama stratejisi ile rastgele bir taban modeli, 4.752 kontrollü sentetik koşu ve gerçek Köyceğiz–Marmaris örneği üzerinde karşılaştırılır. Stratejiler üç düşünce ailesine ayrılır: yerel açgözlü yaklaşımlar (anlık olarak en uygun görünen düğümü korur), küresel yapısal yaklaşımlar (çizgedeki doğal darboğazları arar) ve tek-adım ileri bakışlı politika (motorun kendisini kendi karar süreci içinde benzetir). 4.752 sentetik Delaunay koşusunda tek-adım ileri bakışlı politika ağı ortalama

%27,5'ini yakar; rastgele seçim ise ortalama %70,1'ini yakar — yaklaşık 43 puanlık fark, hareket başına yaklaşık 2 milisaniyelik maliyetle elde edilir. Gerçek Köyceğiz haritasında (101 düğüm, 50.280 ha orman, 172 aktif kenar, hareket başına $k = 2$ itfaiyeci) aynı politika yangını ağın %16,8'inde durdurur. Tüm hat, FastAPI tabanlı arka uç ve React/Leaflet ön uçla sarmalanmış bir web uygulamasında sunulur; bu sayede bir orman planlayıcısı harita üzerine bir sınırlama kutusu çizerek karşılık gelen çizgeyi yaklaşık bir saniyede üretebilir, yangın başlangıç düğümünü seçebilir ve savunulacak düğümlerin sıralı bir listesini, ayrıca topolojiye özgü stratejinin neden seçildiğini açıklayan bir öneri kartıyla birlikte görebilir.

Anahtar Kelimeler: İtfaiyeci Problemi, orman yangını sınırlaması, ağ eniyilemesi, düzlemsel çizgeler, en küçük kesim, ileri bakışlı sezgiseller, karar destek, CBS, ESA WorldCover.

1. Introduction

1.1 The industrial engineering problem

Containing a wildfire is, at its core, a sequential decision problem under tight resource constraints. The fire spreads in time, intervention units are scarce, and at every operational step the planner has to decide which patches of forest to defend first so that the *total* damaged area at the end of the campaign is as small as possible. The question is not how to put a fire out: once the canopy is burning, restoration is on the order of decades, so the only productive lever is *prevention of spread*. That reframing turns a physical phenomenon into a graph-theoretic optimization problem with a clear objective function (total burned vertices), a hard constraint (k vertices defended per turn), and a non-trivial combinatorial structure (the same k units placed at different points yield very different total damage depending on the topology of the surrounding network).

This is the *Firefighter Problem*, introduced by Hartnell in 1995 [1] and known to be NP-hard on general graphs [2], NP-hard even on trees of bounded degree [3], and still APX-hard on planar graphs of maximum degree four [4]. The Firefighter Problem is therefore exactly the kind of question industrial engineering tools are designed for: a polynomial-time exact solution is out of reach, but well-chosen heuristics and structural insights can move the burned fraction by tens of percentage points. Our project takes this framework, adapts it to a real Turkish landscape (Köyceğiz–Marmaris), and packages the resulting heuristics into a decision-support tool that a forestry planner can actually use.

1.2 Identification, analysis, and solution methodology in brief

Identification began with the observation that OGM operations are reactive in nature: the planning lever today is which crew to dispatch to a known fire, not which corridor to defend before the fire arrives. We confirmed the gap with our supervisor and chose the Mediterranean forestry belt (Köyceğiz, Toros Batı, Marmaris) as the concrete pilot region.

Analysis proceeded in three parallel tracks. First, a literature review of the Firefighter Problem

and its planar variants, which gave us the baseline heuristics. Second, the construction of a deterministic simulation engine that mirrors the protect-then-spread mechanics of the formal model. Third, a map-to-graph pipeline that converts ESA WorldCover satellite tiles and OpenStreetMap (OSM) settlement/river data into a planar graph whose vertices are forest patches and whose edges are physically traversable adjacencies after natural barriers (water, bare strips, settlements) are removed.

The solution methodology is a controlled comparison of seven strategies plus a random baseline on 4,752 synthetic instances followed by a single real-map case study. Strategies are grouped into three philosophies, and the best representative in each is highlighted in the recommendation engine. The output is a recommended order in which the planner should defend vertices.

1.3 Expected improvements

Relative to the random baseline (which mimics what an uninformed dispatcher would do), the best strategy reduces average burned fraction from 70.1% to 27.5% across synthetic instances, a 60% relative improvement. Relative to the simplest informed heuristic (max-degree, a natural first thought for a forestry planner), the same strategy reduces the burned fraction from 42.4% to 27.5%, a 35% relative improvement. On the Köyceğiz real-map case the improvement over random is 31.7% → 16.8% (a 47% relative cut). These numbers are not the realistic field expectation, because the model is deterministic and ignores wind; they should be read as relative ordering evidence, not absolute promises.

1.4 Brief summary of conclusions

The most robust finding across all our experiments is that no strategy dominates on every topology, but the ordering between strategies is consistent enough that a topology-aware recommendation engine is worth building. Within that ranking, the one-step lookahead policy — which simulates the engine’s own two-turn behaviour inside the decision — wins on both the synthetic benchmark and the Köyceğiz real-map case, at a cost per decision under 3 ms even on the real graph. A second, narrower result is that reformulating the structural family from “minimise the cost of an arbitrary cut” to “maximise saved-per-cost ratio across multiple candidate cuts” produced a 23% relative reduction in burned fraction (39.4% → 30.3%) against the Vertex Front baseline it replaced. We credit that reformulation directly to the design feedback received from our supervisor on 14 April.

1.5 Contents of the report

Section 2 frames the problem and lists requirements, assumptions and constraints, including the data we gathered for identification and a context diagram. Section 3 reviews the relevant literature, alternative solution approaches, and the methods we eventually chose. Section 4 develops the seven strategies in detail. Section 5 compares them numerically and proposes the recommendation engine. Section 6 covers implementation, integration, and revision cadence. Section 7 closes with the IE tools and methods used, the merits and significance of the design, and a discussion of economic,

environmental, and ethical impacts. The References list and Appendices A–C (test configuration, strategy code map, API endpoints) close the report.

2. Problem Definition, Requirements and Limitations

2.1 What seems to be the problem

The pilot stakeholder is OGM, the Turkish General Directorate of Forestry. Once a fire is reported in a forested district, the operational decision is which patches the available teams should drive to first. The current doctrine is essentially reactive: teams move toward the visible front and try to slow it. There is no analytical layer that says, *given* the topology of the local forest, the village locations and the natural barriers, *defend these two vertices in the next hour and these other two an hour later*. The question we tackle is to build that analytical layer.

Two structural facts make the problem hard. The underlying combinatorial problem (Firefighter on a planar graph) is NP-hard, so an exact optimum cannot be computed on the field timescale, and the resource constraint is binding: typical operational reports suggest two ground teams effectively allocatable to a single front, which we modelled as $k = 2$ vertices protected per turn. A third complication is geometric rather than combinatorial: Mediterranean forests are genuinely long-thin in some places (narrow valley forests along the Köyceğiz lake) and blob-like in others (continuous canopies on the Toros foothills), and the right intervention strategy is not the same in those two regimes.

2.2 What was done to understand the causes

The first three weeks of the project went into understanding *which* features of the problem matter most. The 03.03.2026 meeting established that vertex count, density, and connectivity together drive difficulty levels (low-density narrow corridors are paradoxically easier to defend than high-density blob forests, because the cuts are smaller). The 10.03.2026 meeting compared three graph-generation paradigms and ruled out Erdős–Rényi graphs because their diameter is too small to mimic forest spread realistically; we settled on planar graphs (Delaunay triangulation with edge filters) for synthetic experimentation and on a map-derived planar graph for the real case. The 24.03.2026 meeting refined the vertex-definition rule (vertices represent forest patches and intervention points; lakes and mountains are excluded; edges are dropped where a road, river or bare strip blocks fire transfer). The 14.04.2026 meeting produced the design pivot: the original min-cut strategy was minimising *cut cost*, not *burned vertices*, and a saved/cost ratio reformulation was needed.

We also looked at the operational data that is publicly available. ESA WorldCover (the European Space Agency’s 10 m global land-cover product) gives a forest/shrub/grass/cropland/water/built-up classification of every 10×10 m pixel in Europe and is licensed permissively for non-commercial work. OpenStreetMap supplies village outlines, river polygons, and roads via the Overpass API.

Together they specify, to within a hundred metres, where the forest is, where it is interrupted, and which natural barriers are available. For regional context, the EFFIS annual fire-statistics report for Europe, the Middle East and North Africa [11] and the climate-driven drought analyses of Turco et al. [12] confirm that the Mediterranean rim is in the highest severity band.

2.3 Needs and requirements of the system / customer

ID	Need	Why it matters
R1	Real geography input	The tool must accept a bounding box and produce a graph from the actual landscape, not a stylised abstraction.
R2	Sub-second graph build (cached)	Field use requires that the planner can iterate on bbox/fire origin without losing flow.
R3	Multiple, comparable strategies	Different topologies favour different heuristics; the planner needs to see the comparison and pick.
R4	Per-turn animation	A static “burn percentage” does not communicate <i>where</i> the fire goes; the planner needs the trace.
R5	Recommendation card	The planner is not an OR expert; the tool must explain <i>why</i> a strategy is suggested for the current case.
R6	Reproducibility	Deterministic engine so two planners running the same scenario get the same answer.
R7	Offline operation after first call	Field connectivity is unreliable; tile and Overpass responses are cached on disk.

2.4 Limitations and constraints

Operational. The intervention budget k is small ($k = 2$ in baseline, slider 1–6 in the demo). Protected vertices are immune for the rest of the campaign; once a vertex is burnt, it stays burnt. Time progresses in discrete one-hour steps; within a single step the planner protects k vertices, then the fire spreads simultaneously to every white neighbour of any red vertex.

Modelling. Vertex states are ternary (white, red, green) with no partial damage. Fire spread is deterministic in the baseline: there is no wind, no slope, no fuel-moisture variability. Network topology is static through a campaign; the planner cannot add or remove edges mid-fire.

Computational. The Firefighter Problem is NP-hard on planar graphs. The min-cut family of strategies relies on repeated max-flow computations whose cost scales with the number of vertices; on graphs of more than a few hundred vertices the per-turn cost becomes the binding constraint. We compensated by capping the synthetic graph sizes ($n = 30, 50, 100$) and by giving the recommendation engine the option to skip the heaviest strategies when k is small.

Environmental, social, legal, ethical. ESA WorldCover is freely usable for research; OpenStreetMap is ODbL-licensed and we credit it in the README and the live demo. We deliberately

did not encode population density or settlement-protection priorities into the strategy scores at this stage, because that would require ethical input from OGM about how to value different kinds of asset (a village versus a forest hectare). The recommendation engine therefore stays at the level of *minimising forest hectares burned*; weighting by settlement priority is a flagged future extension.

Geopolitical. All data used is global open-data; the pilot region (Köyceğiz–Marmaris) is on Turkish territory. There is no cross-border data transfer.

2.5 Data gathered and used in the identification phase

- **Köyceğiz–Marmaris bounding box.** N 37.10, S 36.87, E 28.66, W 28.40. Area 589 km².
- **ESA WorldCover 2021 v200.** 10-m resolution land-cover raster; classes 10 (forest) and 20 (shrubland) together produced 50,280 ha of qualifying forest within the bbox.
- **OpenStreetMap Overpass extract.** All natural=water, waterway=river, and place=village|town features inside the bbox. 27 settlements were identified, including Ula, Köyceğiz, Kızılyaka, Marmaris and 23 smaller villages.
- **Synthetic generators.** Delaunay triangulations of uniform random points, long-thin Delaunay (aspect 6×1 , motivated by the 14 April critique), hexagonal lattices, and Delaunay with 18% random edge removal to mimic obstacles. Used for 4,752 simulations across 3 sizes $\times$ 18 seeds $\times$ 11 starting points $\times$ 8 strategies.
- **Wildfire context.** Annual Turkish wildfire bulletins from OGM, EFFIS (European Forest Fire Information System) seasonal reports, and Lloyd’s regional risk map were consulted to confirm that the pilot region is in the top severity band.

2.6 Context diagram

A systemic view of the design problem is given in Figure 1. The system sits between the planner and the data sources: it consumes geographic inputs and configuration parameters from the left, accepts policy and budget constraints from the stakeholder at the top, and produces three flows on the right — a strategy recommendation with a one-line justification, a quantitative performance report, and (through these) operational guidance to the human planner who remains the decision maker.

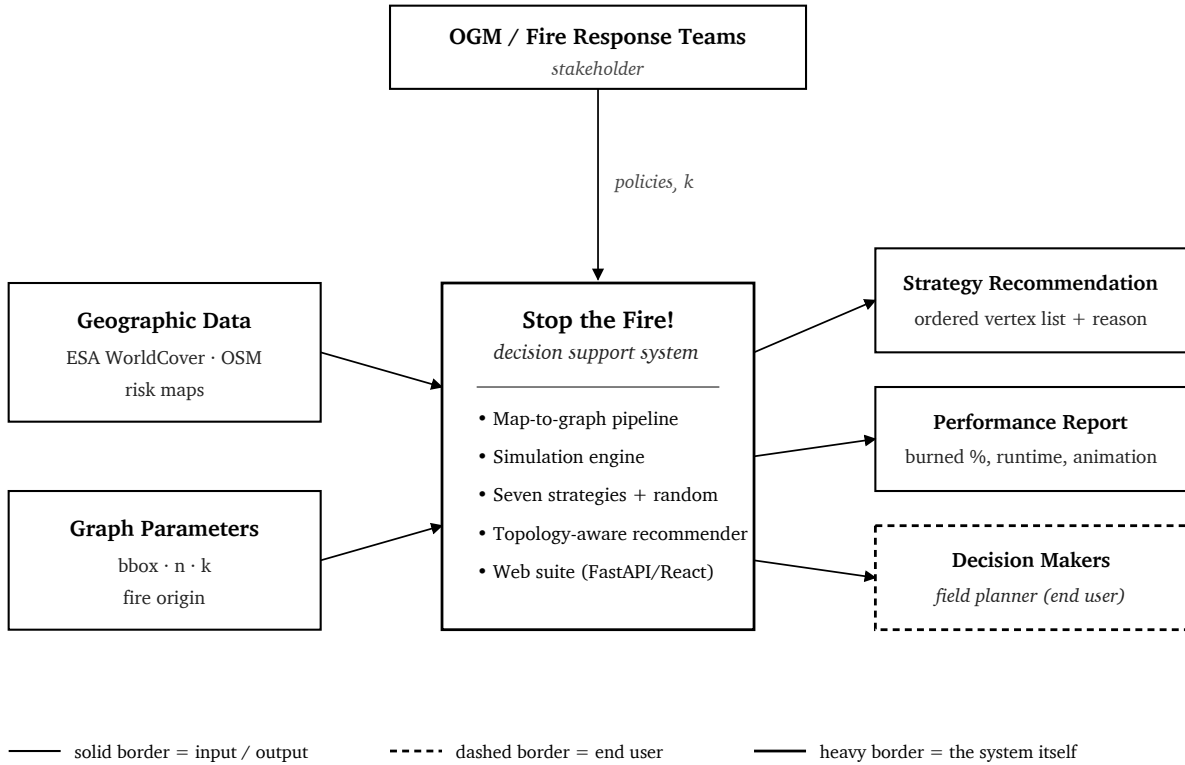


Figure 1: Context diagram of the *Stop the Fire!* decision-support system.

2.7 Performance criteria and potential improvements

The primary criterion is the *burned fraction* at the end of a simulation: the number of red vertices divided by the total number of vertices. Secondary criteria are runtime per turn (we want the recommendation to come back in under a second on a laptop), variance across starting points on the same graph (a strategy that is brilliant on average but disastrous on a quarter of starting points is operationally worse than one that is consistently mediocre), and explainability (the recommendation must be one line of plain language).

Potential improvements identified during identification, in approximately decreasing order of expected impact:

1. **Stochastic spread (wind, slope, fuel moisture).** Will turn deterministic simulations into Monte-Carlo distributions; will change which strategies dominate in marginal cases.
2. **Settlement-priority weighting.** Currently all forest vertices have equal value; weighting vertices near villages would shift the recommendation toward protecting settlement buffers first.
3. **Multi-period budget (carry-over k).** Today's unused intervention capacity vanishes; in the field unused capacity can sometimes be redeployed the next hour.
4. **Live integration with OGM's fire-detection feed.** Today the fire origin is a click; in production it would come from the agency's detection layer.

does *not* weight settlements higher than forest hectares at this stage. The upside is that the tool does not bake in a value-of-life trade-off that the engineering team is not qualified to make; the downside is that in a fire that genuinely threatens a village the unweighted recommendation may not match the operational priority of evacuating people first. Our position is that the tool’s recommendation should be one input among several, and settlement-protection decisions should stay with the human planner, who has both the legitimacy and the local knowledge to weight them. Future versions, after consultation with OGM on the appropriate value weights, can encode settlement priority as a vertex weight in the optimisation objective. A related concern is accountability: when a tool says “defend these two vertices” and a third one burns, the planner faces a difficult conversation. The mitigation we built in is explainability — the recommendation card states *why* a strategy was chosen, and the planner can override it.

References

The references below are the works we drew on for problem framing, algorithm design, data sources and implementation tooling. Citations in the body use the IEEE numeric format [n], referring back to the entries in this list.

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